

Roel-Junior Alejo Viernes

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EDUCATION

University of Greenwich

Sep. 2022 – Jul. 2025

Computer Science BSc (Hons)

Greenwich, London

- **Core Modules:** Data Structures & Algorithms, Object-Oriented Programming (Java), Database Systems (SQL), Machine Learning, Computer Systems Architecture, Advanced Web Technologies
- **Dissertation:** HandTyper – A Real-Time Computer Vision & Low-Latency Gesture Classification Pipeline for Virtual Input Interfaces (Python, Java, OpenCV, Mediapipe)
- **Leadership & Honors:** President of the Computer Science Society (managed team operations, technical workshops, and budgets), Student Representative, Instrumental Ambassador Award

PROJECTS & TECHNICAL CONTRIBUTIONS

HandTyper: Low-Latency Stream Processing Pipeline

Python | Java | OpenCV | Mediapipe

- Architected a real-time computer vision system processing high-frame-rate video feeds, mapping spatial hand landmarks to low-latency virtual keyboard inputs with minimal compute overhead.
- Trained and benchmarked classification models over a dataset of 40,000+ gesture samples, achieving 90% inference accuracy under strict real-time execution constraints.
- Profiled runtime memory usage and stream-handling loops to eliminate processing bottlenecks, ensuring sub-frame response times and zero input drop during continuous data streams.

Sincerely: Cross-Platform Digital Keepsake App

React 19 | TypeScript | Capacitor | Firebase

- Architected a cross-platform Android and PWA app using React 19, TypeScript, and Capacitor 8, building an offline-first mutation sync queue with IndexedDB and Firestore for zero-loss data persistence.
- Implemented an interactive unboxing engine featuring 60 FPS physics-based wax seals, scratch-to-reveal canvas shaders, RevenueCat subscription tiers, and automated 300 DPI QR gift slip generation.

Distributed NFC Telemetry & Web Architecture

Python | JavaScript | HTML/CSS

- Designed and deployed a lightweight, mobile-responsive web platform interfacing with contactless NFC endpoints across physical venues to deliver real-time localized data payloads.
- Engineered backend logging and telemetry scripts to capture and analyze concurrent user interaction events, optimizing endpoint routing and request-response latency for hundreds of users.

EXPERIENCE

Senior UK Student Recruitment Operations Lead

May 2023 – Oct 2025

University of Greenwich

Greenwich, London

- Coordinated a fast-paced operations team of 10+ ambassadors during peak Clearing cycles, triaging high-volume inbound applicant requests across live CRM databases (Dynamics365), telephony, and LiveChat.
- Maintained strict data integrity and resolved complex applicant queries under high-pressure SLA deadlines, optimizing escalation workflows to minimize response latency.

Technical Media & Content Operations Lead

June 2023 – Oct 2025

Faculty of Engineering & Science, University of Greenwich

Greenwich, London

- Collaborated with computer science and engineering faculty researchers to synthesize complex algorithmic and technical concepts into accessible digital technical media and educational demonstrations.
- Analyzed audience engagement analytics and telemetry across multi-channel distribution streams, using data-driven insights to optimize content delivery and increase reach by over 25%.

TECHNICAL SKILLS

Languages : TypeScript, JavaScript (ES6+), Python, Java, SQL, HTML5/CSS3, PowerShell

Frameworks & Cloud : React 19, Capacitor 8 (Android/PWA), Firebase (Firestore, Cloud Storage), Tailwind CSS, OpenCV, Mediapipe

Systems & Tools : Git, GitHub, Linux/Unix CLI, IndexedDB, RevenueCat, VS Code, Dynamics365, Vitest, CI/CD, LaTeX